

Detect Unclassified Files

A DLP feature that identifies files lacking Microsoft Information Protection (MIP) tags and classification labels, enabling organizations to enforce data governance on untagged content.

CloudSec DLP — Feature Documentation

1. MIP Tags (Microsoft Information Protection)

MIP tags are labels embedded within documents by Microsoft Information Protection (or compatible third-party solutions such as Titus / Boldon James). They indicate the sensitivity or classification level of a file — for example, *"Cisco Confidential"*, *"Public"*, or *"Highly Confidential"*.

Where MIP Tags Appear

MIP tags are stored in different locations depending on the file type:

- **Office Documents (Word, Excel, PowerPoint):** Embedded as hidden document properties within the file itself.
- **PDF & RTF files:** Stored in the file's internal metadata.
- **Email (.eml files):** Stored in email headers using the MSIP or Titus standards.

Classification Sources

The system recognises classification metadata from the following sources:

- **Microsoft Information Protection (MSIP)** — Labels applied through Microsoft 365 sensitivity labels.
- **Titus / Boldon James** — Labels applied through the Titus classification suite.

If a file contains metadata from *any* of these sources, it is considered **classified**, regardless of whether the specific tag matches a particular DLP rule.

2. Classification Labels (Platform Labels)

Classification Labels are labels managed by cloud storage platforms such as **Box**, **Office 365**, and **Google Drive**. Unlike MIP tags (which are embedded inside the file content), these labels are applied and managed through the platform itself.

Supported Label Platforms

- **Box** — Box Classifications and metadata labels
- **Office 365** — Sensitivity labels from Microsoft Purview
- **Google Drive** — Google Drive labels

Key distinction: MIP tags live *inside* the file content (document properties / email headers). Classification labels are *external* metadata managed by the cloud platform. Both are indicators that a file has been classified.

3. What Are Unclassified Files?

An **unclassified file** is a file that has *no* classification metadata associated with it — neither platform labels nor embedded MIP tags.

A file is considered unclassified when:

1. **No platform labels** — The cloud storage platform (Box, Office 365, Google Drive) reports zero classification labels for the file.
2. **No MIP tag metadata** — The file content does not contain any embedded classification properties from Microsoft or Titus.

Why unclassified files matter

Organizations that mandate data classification need visibility into files that have *not* been tagged. Unclassified files represent a compliance gap — they may contain sensitive data without the proper labels that trigger downstream protection policies. Detecting these files enables security teams to take corrective action, such as notifying the file owner or enforcing classification.

4. Feature Overview

The **Detect Unclassified Files** feature is a DLP (Data Loss Prevention) capability that flags files which lack any classification metadata. It works alongside MIP tag matching and data classification scanning to provide comprehensive coverage.

What the feature does

- Adds an **"Unclassified Files"** toggle to DLP policy rules under the Files Control section.
- When enabled on a rule, the system creates a **DLP incident** if a file has no classification labels and no embedded MIP tag metadata.
- Can optionally be combined with data classifiers — a rule can detect unclassified files *and* inspect the file content for sensitive patterns (e.g., credit card numbers, social security numbers).

Important: A rule **cannot** have both "MIP and Titus Labels" and "Unclassified Files" enabled at the same time. These are two separate, mutually exclusive options. A rule either looks for specific MIP tags *or* detects the absence of any classification — never both.

5. Policy Rule Configuration — UI Changes

A new "**Unclassified Files**" option has been added to the **Files Control** section of the DLP policy rule configuration screen. This sits alongside the existing file control options.

Files Control

Include filters for the files that this rule will search for when inspecting document properties.

MIP and Titus Labels

Enable to scan files with Microsoft Information Protection labels added in MS365

☐ Disabled

File Size Filter

Enable to scan files with specific size or within a size range.

☐ Disabled

File Type Filter

Enable to scan specific file types such as pdf, docx, csv, etc.

☐ Disabled

Unclassified Files

Enable to scan and log files with no labels at all.

☐ Disabled

Files Control section in the DLP policy rule configuration screen

Files Control Options

The Files Control section now includes the following options when creating or editing a DLP policy rule:

1 **MIP and Titus Labels**

Enable to scan files with Microsoft Information Protection labels added in MS365. When turned on, you can specify which MIP tag names to look for (e.g., "Cisco Confidential").

2 **File Size Filter**

Enable to scan files with a specific size or within a size range.

3 **File Type Filter**

Enable to scan specific file types such as PDF, DOCX, CSV, etc.

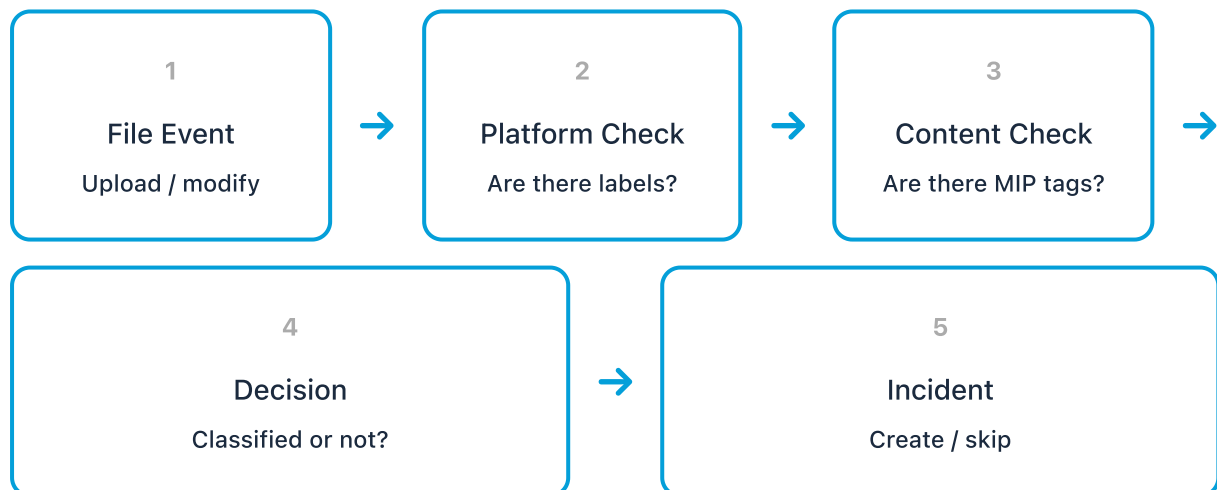
4 **Unclassified Files (New)**

Enable to scan and log files with no labels at all. When turned on, the system will create an incident for any file that has no classification — no MIP tags, no Titus labels, and no platform labels. This option is **mutually exclusive** with MIP and Titus Labels — you cannot enable both on the same rule.

How to enable: Navigate to the DLP policy rule editor, scroll to the **Files Control** section, and toggle the **"Unclassified Files"** switch to "Enabled". Save the rule. Any file processed by this rule that has no classification metadata will generate a DLP incident.

6. How It Works

When a file is uploaded or modified in a connected cloud platform, the system processes it through the following steps:



Step-by-Step

1. **File event detected:** A user uploads, creates, or modifies a file in a connected cloud platform (Box, Office 365, Google Drive, ServiceNow, Dropbox, etc.).
2. **Platform label check:** The system checks whether the cloud platform has any classification labels assigned to the file.
3. **Content-level MIP tag check:** For supported file types, the system opens the file and inspects its internal metadata for MIP tags or Titus classification properties.

4. **Classification decision:** If the file has *no* platform labels *and no* embedded MIP/Titus tags, it is determined to be **unclassified**.
5. **Incident creation:** If the DLP rule has "Unclassified Files" enabled and the file is unclassified, a DLP incident is created. If the file has any form of classification, no incident is generated for that rule.

Key Behavioral Rules

Rules with "Unclassified Files" enabled

- Incidents are created when the file has **no classification metadata at all** (no platform labels, no MIP tags, no Titus labels).
- If the file is classified (has any MIP tag or platform label), the rule does **not** create an incident.
- If a data classifier is also configured on the rule, classified files are still excluded — the classification check takes priority.

Rules with "MIP and Titus Labels" enabled

- Incidents are created only when the file's embedded MIP tag matches one of the specific tags configured on the rule.
- If the MIP tag doesn't match, no incident is created — even if a data classifier matches the file content.

Rules with only data classifiers (no MIP tags, no Unclassified Files)

- Incidents are created purely based on content matching. The file's classification status is irrelevant.

In summary: MIP tag rules look for specific labels. Unclassified Files rules look for the *absence* of any labels. Plain data classifier rules are indifferent to classification status.

7. Scenarios — Complete Matrix

The following table shows every combination of rule settings, file classification state, and data classifier outcome. "MIP and Titus Labels" and "Unclassified Files" are

mutually exclusive — no scenario has both enabled on the same rule.

#	Rule: MIP Tags	Rule: Detect Unclassified	Data Classification	File: MIP *
1	N/A	FALSE	Hello	N/A
2	N/A	FALSE	Hello	Cisco Confident
3	N/A	FALSE	Hello	N/A
4	N/A	FALSE	Hello	Cisco Confident
5	Cisco Confidential	FALSE	N/A	N/A
6	Cisco Confidential	FALSE	N/A	Cisco Confident
7	Cisco Confidential	FALSE	N/A	N/A
8	Cisco Confidential	FALSE	N/A	Cisco Confident
9	Cisco Confidential	FALSE	Hello	N/A
10	Cisco Confidential	FALSE	Hello	Cisco Confident
11	Cisco Confidential	FALSE	Hello	N/A
12	Cisco Confidential	FALSE	Hello	Cisco Confident
13	N/A	TRUE	N/A	N/A

#	Rule: MIP Tags	Rule: Detect Unclassified	Data Classification	File: MIP
14	N/A	TRUE	N/A	Cisco Confident
15	N/A	TRUE	N/A	N/A
16	N/A	TRUE	N/A	Cisco Confident
17	N/A	TRUE	Hello	N/A
18	N/A	TRUE	Hello	Cisco Confident
19	N/A	TRUE	Hello	N/A
20	N/A	TRUE	Hello	Cisco Confident

How to read the table:

- **Rule: MIP Tags** — The MIP tag name configured on the rule (N/A means none configured).
- **Rule: Detect Unclassified** — Whether "Unclassified Files" is enabled on the rule.
- **Data Classification** — The content pattern/classifier configured on the rule (N/A means none).
- **File: MIP Tag** — The MIP tag actually present on the file being scanned (N/A means the file has no MIP tag).
- **Data Classifier** — Whether the file content matched the configured pattern (N/A means no match or no classifier).
- **Expected Result** — Whether a DLP incident is created or not.

Key takeaway: When "Detect Unclassified" is **TRUE** (blue-highlighted rows 13–20), the presence of any classification on the file *always* prevents an incident —

even if a data classifier matches. Conversely, any file without classification will trigger an incident for that rule.

8. Supported File Types

All file types are supported by the Detect Unclassified Files feature. The system evaluates every file regardless of its format or extension.